

APPENDIX TO "THE METHOD OF MODERATION"

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ABSTRACT

This appendix provides detailed mathematical derivations and technical results supporting the Method of Moderation. Topics include: value function transformations and their relationship to the inverse value function; explicit formulas for minimal and maximal marginal propensities to consume; cusp point calculations for tighter upper bounds; Hermite interpolation slope formulas and MPC derivations; patience conditions ensuring well-defined solutions; and extensions to stochastic returns with explicit formulas for portfolio problems.

Keywords Dynamic Stochastic Optimization, Consumption-Saving Models, Numerical Methods

1 APPENDIX: MATHEMATICAL DETAILS

1.1 Patience Conditions Details

The patience conditions listed in the main text have clear economic interpretations. The FVAC $0 < \beta G^{1-\rho} \mathbf{E}[\psi^{1-\rho}] < 1$ ensures that autarky (saving nothing, consuming all income each period) yields finite expected discounted utility, guaranteeing the consumer values resources. The AIC $\Phi < 1$ prevents indefinite consumption deferral by ensuring the marginal utility of current consumption exceeds the discounted marginal utility of future consumption under certainty. The RIC $\Phi/R < 1$ ensures asset growth is slower than the patience-adjusted discount rate, preventing unbounded wealth accumulation. The GIC $\Phi/G < 1$ ensures consumption grows slower than permanent income, establishing a target wealth ratio. The FHC $G/R < 1$ ensures the present value of future labor income is finite. Together, these conditions partition parameter space into regions with qualitatively different behavior: buffer-stock saving with a target wealth ratio (all conditions hold), perpetual borrowing (AIC fails), or unbounded wealth growth (GIC fails but RIC holds) [1, 2, 3].

1.2 Human Wealth Formulas

The optimist's human wealth (assuming $\xi_{t+n} = 1 \forall n > 0$) can be computed three ways: backward recursion $\bar{h}_T = 0$, $\bar{h}_t = (G/R)(1 + \bar{h}_{t+1})$; forward sum $\bar{h}_t = \sum_{n=1}^{T-t} (G/R)^n$; or infinite-horizon $\bar{h} = G/(R - G)$ when $R > G$. With $G = 1$, $\bar{h} = 1/(R - 1)$.

The pessimist's human wealth (assuming $\xi_{t+n} = \xi \forall n > 0$) follows similarly: backward recursion $\underline{h}_T = 0$, $\underline{h}_t = (G/R)(\xi + \underline{h}_{t+1})$; forward sum $\underline{h}_t = \xi \sum_{n=1}^{T-t} (G/R)^n$; or infinite-horizon $\underline{h} = \xi G/(R - G)$. When $\xi = 0$ (unemployment), $\underline{h} = 0$.

1.3 Marginal Propensity to Consume Formulas

The minimal MPC (perfect foresight consumer with horizon $T - t$) has three forms [4]: backward recursion $\underline{\kappa}_t = \underline{\kappa}_{t+1}/(\underline{\kappa}_{t+1} + \Phi/R)$ with $\underline{\kappa}_T = 1$; forward sum $\underline{\kappa}_t = (\sum_{n=0}^{T-t} (\Phi/R)^n)^{-1}$; or infinite-horizon $\underline{\kappa} = 1 - \Phi/R = 1 - (R\beta)^{1/\rho}/R$.

The maximal MPC [5] satisfies backward recursion $\bar{\kappa}_t = \bar{\kappa}_{t+1}/(\bar{\kappa}_{t+1} + \wp^{1/\rho}\Phi/R)$ with $\bar{\kappa}_T = 1$; forward sum $\bar{\kappa}_t = (\sum_{n=0}^{T-t} (\wp^{1/\rho}\Phi/R)^n)^{-1}$; or infinite-horizon $\bar{\kappa} = 1 - \wp^{1/\rho}(\Phi/R)$.

1.4 Cusp Point Calculation

The two upper bounds intersect at the cusp point m^* where

$$\begin{aligned} (\Delta m^* + \Delta h) \underline{\kappa} &= \bar{\kappa} \Delta m^* \\ \Delta m^* &= \frac{\underline{\kappa} \Delta h}{\bar{\kappa} - \underline{\kappa}} \\ m^* &= -\underline{h} + \frac{\underline{\kappa} (\bar{h} - \underline{h})}{\bar{\kappa} - \underline{\kappa}}, \end{aligned} \quad (1)$$

where $\Delta m^* \equiv m^* - \underline{m} > 0$ since $\bar{\kappa} > \underline{\kappa}$. For $m \in (\underline{m}, m^*]$, the tighter upper bound yields

$$\begin{aligned} \Delta m \underline{\kappa} &< \hat{c}(\underline{m} + \Delta m) &< \bar{\kappa} \Delta m \\ 0 &< \hat{c}(\underline{m} + \Delta m) - \Delta m \underline{\kappa} &< \Delta m (\bar{\kappa} - \underline{\kappa}) \\ 0 &< \left(\frac{\hat{c}(\underline{m} + \Delta m) - \Delta m \underline{\kappa}}{\Delta m (\bar{\kappa} - \underline{\kappa})} \right) &< 1. \end{aligned} \quad (2)$$

This motivates the definition of the low-resource moderation ratio as in (??).

1.5 Value Function Derivation

Under perfect foresight, consumption grows at constant rate equal to the absolute patience factor Φ : $c_{t+n} =$

$c_t \Phi^n$. The present discounted value of consumption satisfies $\text{PDV}_t^T(c) = \sum_{n=0}^{T-t} \beta^n c_t \Phi^n = c_t \sum_{n=0}^{T-t} (\Phi/R)^n$, where we use $\beta \Phi^{1-\rho} = \Phi/R$. Dividing by consumption yields the PDV-to-consumption ratio $C_t^T = \text{PDV}_t^T(c)/c_t = \sum_{n=0}^{T-t} (\Phi/R)^n = \underline{\kappa}_t^{-1}$, which is unchanged for normalized variables. Defining $\bar{C} \equiv \lim_{T \rightarrow \infty} C_t^T$, this yields the key identity $\bar{C} = \underline{\kappa}^{-1}$, connecting the infinite-horizon PDV-to-consumption ratio to the minimal MPC.

The optimist's value function satisfies

$$\begin{aligned} \bar{v}_{T-1}(m_{T-1}) &\equiv \mathbf{u}(c_{T-1}) + \beta \mathbf{u}(c_T) \\ &= \mathbf{u}(c_{T-1}) (1 + \beta \Phi^{1-\rho}) \\ &= \mathbf{u}(c_{T-1}) (1 + \Phi/R) \\ &= \mathbf{u}(c_{T-1}) C_{T-1}^T \end{aligned} \quad (3)$$

The infinite horizon expression becomes

$$\begin{aligned} \bar{v}(m) &= \mathbf{u}(\bar{c}(m)) \bar{C} \\ &= \mathbf{u}(\bar{c}(m)) \underline{\kappa}^{-1} \\ &= \mathbf{u}((\Delta m + \Delta h) \underline{\kappa}) \underline{\kappa}^{-1} \\ &= \mathbf{u}(\Delta m + \Delta h) \underline{\kappa}^{-\rho}. \end{aligned} \quad (4)$$

This can be transformed as

$$\begin{aligned} \bar{\lambda} &\equiv ((1 - \rho) \bar{v})^{1/(1-\rho)} \\ &= c \bar{C}^{1/(1-\rho)} \\ &= (\Delta m + \Delta h) \underline{\kappa}^{-\rho/(1-\rho)}. \end{aligned} \quad (5)$$

For the realist's problem, we define $\hat{\lambda} = ((1 - \rho) \hat{v}(m))^{1/(1-\rho)}$. Using the bounds $\underline{\lambda} < \hat{\lambda} < \bar{\lambda}$, we define

$$\hat{\Omega}(\mu) = \left(\frac{\hat{\lambda}(\underline{m} + e^\mu) - \underline{\lambda}(\underline{m} + e^\mu)}{\Delta h \underline{\kappa} \bar{C}^{1/(1-\rho)}} \right) \quad (6)$$

and the logit-transformed counterpart:

$$\begin{aligned} \hat{X}(\mu) &= \log \left(\frac{\hat{\Omega}(\mu)}{1 - \hat{\Omega}(\mu)} \right) \\ &= \log(\hat{\Omega}(\mu)) - \log(1 - \hat{\Omega}(\mu)) \end{aligned} \quad (7)$$

Inverting these approximations yields

$$\hat{\lambda} = \underline{\lambda} + \overbrace{\left(\frac{1}{1 + \exp(-\hat{X})} \right)}^{=\hat{\Omega}} \Delta h \underline{\kappa} \bar{C}^{1/(1-\rho)} \quad (8)$$

from which the value function approximation is $\hat{v} = \mathbf{u}(\hat{\lambda})$.

1.6 I.I.D. Stochastic Returns: MPC derivation

The fact that a linear consumption function with an $\text{MPC} = 1 - (\beta \mathbf{E}[\mathbf{R}^{1-\rho}])^{1/\rho}$ satisfies the Euler equation with i.i.d. returns

and no labor income can be derived by the method of undetermined coefficients. In particular, assume that $\bar{c}(m) = m \underline{\kappa}$, with a time-independent MPC $\underline{\kappa}$ to be determined. Substituting this into the Euler equation, we have

$$\begin{aligned} 1 &= \beta \mathbf{E}_t \left[\mathbf{R}_{t+1} \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \right] \\ &= \beta \mathbf{E}_t \left[\mathbf{R}_{t+1} \left(\frac{m_{t+1}}{m_t} \right)^{-\rho} \right] \end{aligned} \quad (9)$$

where the second equality uses the assumed form of the consumption function. Since there is no labor income, $m_{t+1} = \mathbf{R}_{t+1}(m_t - c_t)$. Substituting this into the above we obtain

$$1 = \beta \mathbf{E}_t \left[\mathbf{R}_{t+1} (\mathbf{R}_{t+1} (1 - \underline{\kappa}))^{-\rho} \right] \quad (10)$$

Solving for $\underline{\kappa}$ and recalling that returns are i.i.d. gives $\underline{\kappa} = 1 - (\beta \mathbf{E}[\mathbf{R}^{1-\rho}])^{1/\rho}$.

In the particular case of lognormal returns, the MPC can be written in closed form. The moment generating function (MGF) for lognormal returns provides the key formula. For $\log \mathbf{R} \sim \mathcal{N}(\mu, \sigma^2)$, the MGF is $\mathbf{E}[e^{sX}] = \exp(\mu s + \sigma^2 s^2/2)$ where $X = \log \mathbf{R}$. Setting $s = 1 - \rho$ and $\mu = r + \pi - \sigma_r^2/2$ yields³

$$\mathbf{E}[\mathbf{R}^{1-\rho}] = \exp \left((1 - \rho) \left(r + \pi - \frac{\sigma_r^2}{2} \right) + \frac{(1 - \rho)^2 \sigma_r^2}{2} \right). \quad (11)$$

Simplifying the variance terms: $(1 - \rho)^2 \sigma_r^2/2 - (1 - \rho) \sigma_r^2/2 = (1 - \rho)[(1 - \rho) - 1] \sigma_r^2/2 = -\rho(1 - \rho) \sigma_r^2/2$, giving the final form

$$\mathbf{E}[\mathbf{R}^{1-\rho}] = \exp \left((1 - \rho) (r + \pi - \rho \sigma_r^2/2) \right). \quad (12)$$

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³Here we can interpret π as the risk premium, that is, the additional average return from holding a risky asset compared to the risk-free rate r . Adjusting the average log return by the asset volatility ensures that increasing σ_r^2 constitutes a mean-preserving spread of the level of return.

[//llorracc.github.io/BufferStockTheory/BufferStockTheory.pdf](https://llorracc.github.io/BufferStockTheory/BufferStockTheory.pdf).

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